DSAA3073 · Theories in Data Science

Choose feasible decisions as losses change

Learning Sheet · Lecture 11

Online convex optimization, projection, and regularization

Compare continuous decisions on one sequence

An online learner chooses before seeing the current loss. With finitely many experts, the learner can keep one weight for each expert. A decision can instead be a real vector, such as the parameters of a predictor. The vector must belong to an allowed region. Keeping one weight for every possible point is then impossible. We need to use the geometry of the region and the shape of the losses.

LESSON BACKBONE

How can we choose a feasible point before seeing its loss and still stay close to one good fixed decision?

We will use convexity to compare points, projection to keep updates feasible, and two different arguments to bound the accumulated loss: changes in distance and stability of regularized decisions.

A finite grid is one possible approximation, but its size can grow quickly. A product grid with ten choices in each of ten coordinates has 10^{10} points. Constraints can remove many of these points, so this count is not the number of feasible points in every constrained problem. A vector has only ten coordinates, while listing a fine grid can require far more entries. A gradient offers information without listing all candidate points.

Specify the decision and its loss

A set is **convex** if the line segment joining any two of its points stays in the set. Thus an average of feasible decisions remains feasible. Intervals, boxes and Euclidean balls are examples. Here "feasible" means that the decision satisfies the specified constraints.

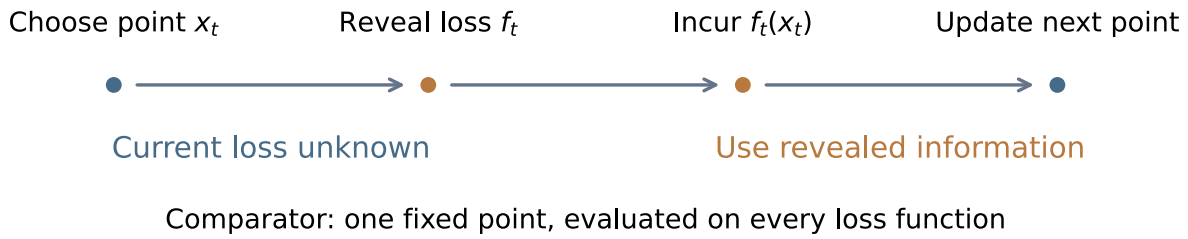
Definition: Online convex optimization

Let $\mathcal{K} \subseteq \mathbb{R}^d$ be a nonempty compact convex decision set. Compact means closed and bounded in this finite-dimensional setting. On round t:

1. Choose a point $x_t \in \mathcal{K}$ using information already available.
2. Observe a convex loss function $f_t : \mathcal{K} \to \mathbb{R}$ and incur the scalar loss $f_t(x_t)$.
3. Use the revealed information to choose the next point.

We use losses that are differentiable on a neighborhood of $\mathcal{K}$. The gradient at the played point is $g_t = \nabla f_t(x_t)$. The functions may change arbitrarily between rounds; they need not be independent samples from a distribution.

The set is fixed throughout the sequence. The point changes each round. The function f_t assigns a loss to every feasible point, while $f_t(x_t)$ is one number. This distinction matters: observing one gradient does not reveal the loss function of a future round.



The current point is chosen before the current loss function is revealed. The arrows carry information forward. The comparator below the timeline is one fixed point, not a separate best point on each round.

We assume full loss-function feedback in the protocol. Some algorithms below need only the gradient at the played point. A method that minimizes all past loss functions can use the fuller feedback. Neither may use the current function before choosing the current point.

Keep the comparator fixed

The same comparison used for expert advice now ranges over points in $\mathcal{K}$. Add each candidate point's losses over the complete sequence, then choose the point with the smallest sum.

Definition: Regret for continuous decisions

After T rounds, regret is

$$R_T = \sum_{t=1}^{T} f_t(x_t) - \min_{u \in \mathcal{K}} \sum_{t=1}^{T} f_t(u).$$

The minimum uses one fixed feasible point u on every round. Continuity of the losses and compactness of $\mathcal{K}$ ensure that a minimizer exists. Without an attainment assumption, the definition uses an infimum in place of the minimum.

The comparator is selected after the sequence is known. The learner does not need to identify it while playing. To prove a regret bound, it is enough to compare the learner with **every** fixed $u \in \mathcal{K}$ and then choose a minimizer at the end.

For example, suppose the losses on $[-1, 1]$ are $f_1(x) = x$ and $f_2(x) = -x$. Every fixed point has total loss zero. Choosing -1 on round 1 and 1 on round 2 would give total loss -2, but those are two different decisions. Their sum is not the fixed-comparator value. Losses in this continuous setting may be negative; the coming bounds control gradients and distances rather than assuming losses lie in $[0, 1]$.

Use convexity to compare with an unknown point

A differentiable convex function lies above each of its tangent planes. At the played point, this means

$$f_t(u) \geq f_t(x_t) + g_t^T (u - x_t)$$

for every feasible u.

Rearranging gives the comparison we need:

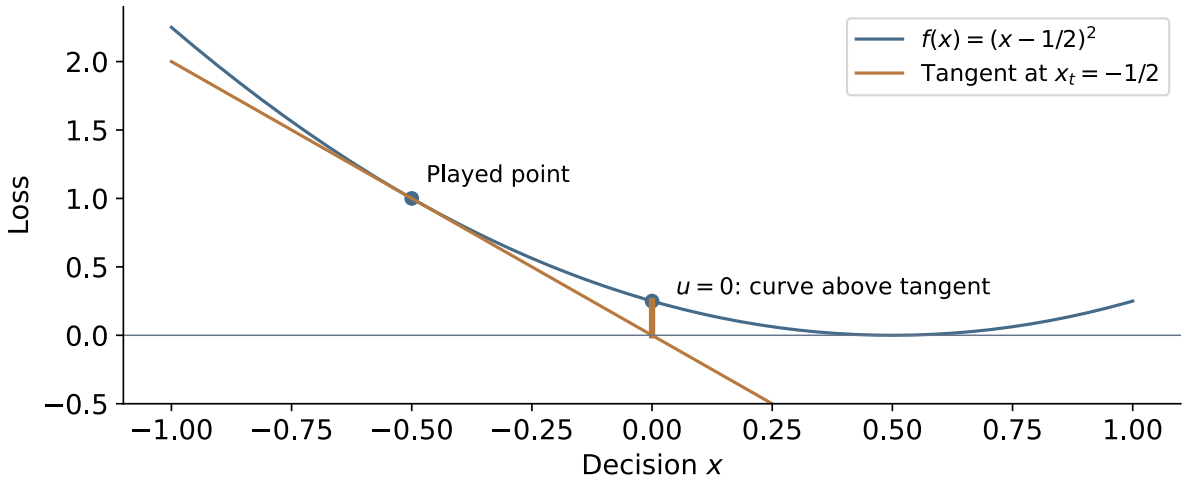
Result: Convex linearization

For a differentiable convex loss and any $u \in \mathcal{K}$,

$$f_t(x_t) - f_t(u) \le g_t^T (x_t - u).$$

The left side is the actual loss difference. The right side is a linear expression involving the observed gradient and the displacement from the comparator to the played point.

This is an upper bound on a loss difference, not a prediction of tomorrow's loss. It also does not say that the gradient points toward the best fixed point. A gradient describes local increase of the current loss; the comparator depends on the entire sequence.



At $x_t = -\frac{1}{2}$, the quadratic has value 1 and slope -2. Its tangent lies below the curve at $u = 0$. Subtracting the curve value at u therefore gives a smaller gap than subtracting the tangent value.

Worked example: Read the tangent comparison

For $f(x) = \left(x - \frac{1}{2}\right)^2$, take $x_t = -\frac{1}{2}$ and $u = 0$. The supplied values are

$$f(x_t) = 1, \quad f(u) = \frac{1}{4}, \quad g_t = -2.$$

The actual gap is $\frac{3}{4}$. Its linear upper bound is $(-2)\left(-\frac{1}{2} - 0\right) = 1$. Thus $\frac{3}{4} \le 1$. The upper bound need not be an equality; its benefit is that it uses the gradient at the point actually played.

We have replaced a possibly curved loss difference with an inner product. We still need a decision rule that controls the sum of these inner products. A gradient step is a natural candidate, provided its result remains a legal decision.

Keep the next decision feasible

Moving opposite the current gradient can leave the allowed region. For a constrained predictor, an out-of-region parameter is not an admissible next decision. The first requirement is therefore

to return the proposal to $\mathcal{K}$. We will choose the closest feasible point and show why that choice is useful for the comparison proof.

Definition: Euclidean projection

For a nonempty closed convex set $\mathcal{K}$ and any $y \in \mathbb{R}^d$, define

$$\Pi_{\mathcal{K}}(y) = \arg\min_{x \in \mathcal{K}} \|x - y\|_2.$$

The closest point exists and is unique. Minimizing the squared distance gives the same point. If y is already feasible, its projection is y itself.

Existence follows from closedness and the growth of the distance far from y. Uniqueness follows from strict convexity of squared Euclidean distance on a convex set: two distinct minimizing points would have a midpoint with smaller squared distance. The target set must be included in the notation because projecting the same proposal onto different sets can give different answers.

See what the projection changes

For an interval, projection clips a number to its nearest endpoint if it lies outside. For a box, apply this clipping separately in each coordinate. The following supplied two-dimensional step makes both the proposal and the correction visible.

Worked example: A proposal outside a box

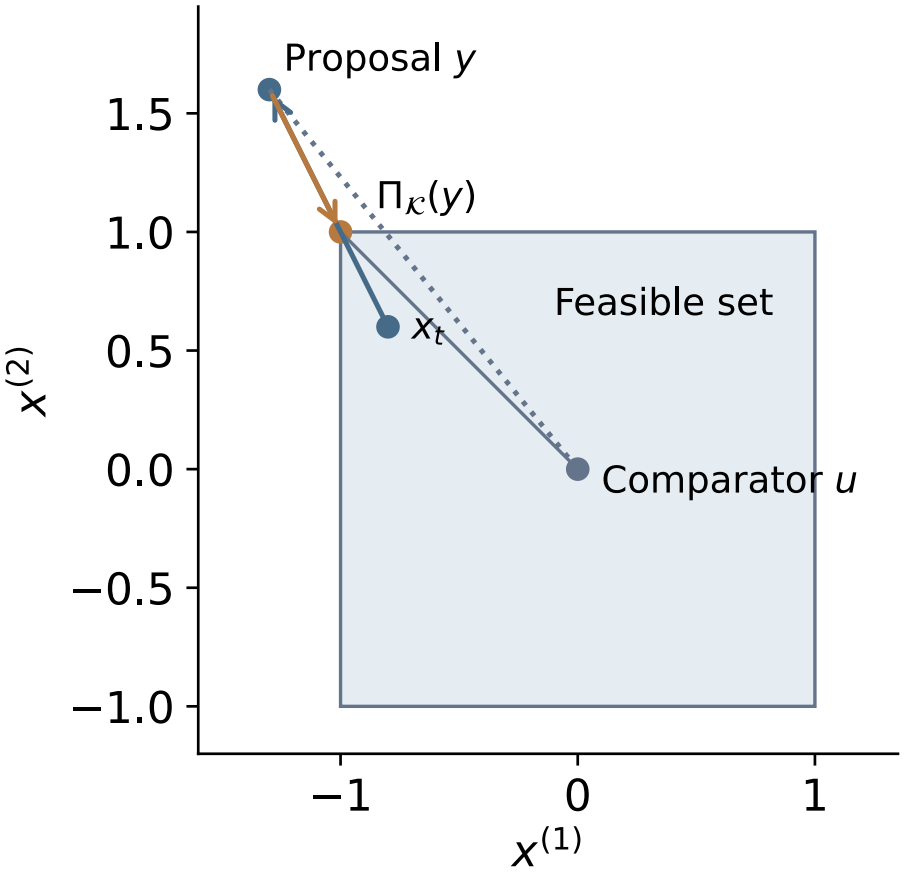
Let $\mathcal{K} = [-1, 1]^2$, $x_t = (-0.8, 0.6)$, $g_t = (1, -2)$ and $\eta = \frac{1}{2}$. The raw gradient proposal is

$$y = x_t - \eta g_t = (-1.3, 1.6).$$

It violates both coordinate bounds. Clipping gives

$$\Pi_{\mathcal{K}}(y) = (-1, 1).$$

The correction changes the proposed step so that the next decision belongs to the original set. It does not change that set or the current loss already incurred.



The blue arrow takes the gradient step; the amber arrow returns the proposal to the box. The two lines to $u = (0, 0)$ compare distances before and after projection. Supplied squared distances are 4.25 before projection and 2 afterward.

Projection is not always coordinatewise clipping. On a ball $\mathcal{K} = \{x : \|x\|_2 \le B\}$ with $B > 0$, the formula is

$$\Pi_{\mathcal{K}}(y) = \begin{cases} y & \text{if } \|y\|_2 \le B \\ \frac{By}{\|y\|_2} & \text{if } \|y\|_2 > B. \end{cases}$$

An outside point is moved along its radial line to the boundary. Box and ball projections can be computed with a number of operations proportional to dimension. More general convex sets may require a separate optimization procedure. The geometric guarantee below does not make that computation free.

Prove the distance property

Let $p = \Pi_{\mathcal{K}}(y)$ and choose any feasible comparator u. Every point $p + s(u - p)$ with $0 \le s \le 1$ is feasible. Since p minimizes squared distance to y, moving a small positive amount along this segment cannot initially decrease that distance.

Differentiate the squared distance along the segment at $s = 0$:

$$\frac{d}{ds}\|p + s(u - p) - y\|_2^2 \mid_{s=0} = 2(p - y)^T(u - p) \ge 0.$$

Thus $(y-p)^T(u-p) \leq 0$. The sign is important: the vector from p to y has a nonpositive inner product with every direction from p into the feasible set.

Expand the distance from the proposal to the comparator:

$$\|y-u\|_2^2 = \|y-p\|_2^2 + \|p-u\|_2^2 + 2(y-p)^T(p-u).$$

The last inner product is nonnegative by the preceding inequality. Dropping it and the nonnegative term $\|y-p\|_2^2$ gives the desired result.

Result: Projection preserves the comparator-distance bound

For every $y \in \mathbb{R}^d$ and $u \in \mathcal{K}$,

$$\|\Pi_{\mathcal{K}}(y) - u\|_2^2 \leq \|y-u\|_2^2.$$

Projection does not increase distance to any feasible comparator. It also restores feasibility. These are different reasons for using the same operation.

This statement concerns the proposal and its projection. It does not say that the complete gradient step makes the new decision closer to u than the previous decision was. That distance may increase on some rounds. Our proof will control its accumulated changes instead.

Play, observe, step, project

The algorithm can now be stated without an unexplained operation. Play the current point. After seeing its loss, move opposite the current gradient. Project that proposal back into the allowed region, and use the result on the next round.

Definition: Online gradient descent

Choose $x_1 \in \mathcal{K}$ and positive step sizes η_t. On round t, play x_t, observe f_t and set $g_t = \nabla f_t(x_t)$. Then update

$$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t g_t).$$

The gradient from round t influences round $t+1$. The current point was chosen before that gradient was available.

We will prove a bound using a constant step size $\eta_t = \eta > 0$. A time-dependent step is allowed in the algorithm's definition, but it needs its own treatment when summing the distance inequalities.

Worked example: Follow one fixed loss sequence

Take $\mathcal{K} = [-1, 1]$, $x_1 = 0$, $\eta = \frac{1}{2}$ and linear losses $f_t(x) = g_t x$. The coefficients and computed decisions are supplied:

Round	g_t	Played x_t	Loss	Next point
1	$\frac{1}{2}$	0	0	$-\frac{1}{4}$
2	-1	$-\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
3	1	$\frac{1}{4}$	$\frac{1}{4}$	$-\frac{1}{4}$
4	-1	$-\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

All proposals happen to lie in the interval, so projection leaves them unchanged. The learner's total loss is $\frac{3}{4}$. The sum of the loss functions is $-\frac{x}{2}$, minimized at $u = 1$ with value $-\frac{1}{2}$. Regret is therefore $\frac{3}{4} - \left(-\frac{1}{2}\right) = \frac{5}{4}$.

The decisions move left and right because the losses change. This does not by itself establish either a good or a bad regret rate. A finite trace illustrates the rule; a theorem must compare every allowed sequence with its own best fixed point.

Turn distance changes into a regret bound

Fix an arbitrary $u \in \mathcal{K}$ for the entire proof. We do not assume that the learner knows this point. Convex linearization bounds the round's loss difference by $g_t^T(x_t - u)$. The update will turn that inner product into a difference of squared distances plus a term involving the gradient size.

Isolate the comparator inner product

Apply the projection result to the raw proposal $x_t - \eta g_t$:

$$\|x_{t+1} - u\|_2^2 \le \|x_t - \eta g_t - u\|_2^2.$$

Expanding the square on the right gives

$$\|x_{t+1} - u\|_2^2 \le \|x_t - u\|_2^2 - 2\eta g_t^T(x_t - u) + \eta^2 \|g_t\|_2^2.$$

Move the inner-product term to the left and divide by positive 2η. The direction of the inequality is preserved:

$$g_t^T(x_t - u) \le \frac{\|x_t - u\|_2^2 - \|x_{t+1} - u\|_2^2}{2\eta} + \frac{\eta}{2}\|g_t\|_2^2.$$

Finally insert the convexity bound on the actual loss difference.

Result: One-round OGD comparison

For every fixed $u \in \mathcal{K}$ and every round,

$$f_t(x_t) - f_t(u) \leq \frac{\|x_t - u\|_2^2 - \|x_{t+1} - u\|_2^2}{2\eta} + \frac{\eta}{2}\|g_t\|_2^2.$$

The first term measures a change in squared distance. The second accounts for the size of the step through the squared gradient norm.

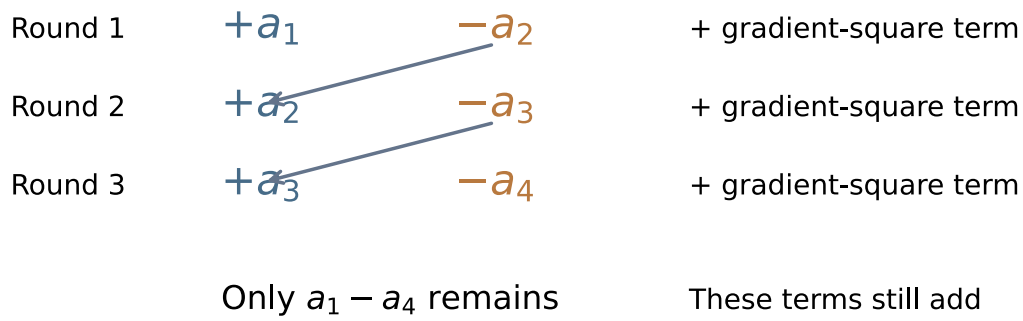
The distance term can be negative. If the new point is farther from u, the difference of squared distances is negative, and the remaining term still contributes to the upper bound. No probabilistic noise assumption was used. The gradient-size term is present even for a fully deterministic sequence.

Sum before replacing the distances by bounds

For a short view of the cancellation, temporarily write $a_t = \|x_t - u\|_2^2$. Over three rounds,

$$(a_1 - a_2) + (a_2 - a_3) + (a_3 - a_4) = a_1 - a_4.$$

The same cancellation occurs over any number of rounds. This is called **telescoping**: intermediate terms appear once positively and once negatively.



The symbol a_t here means squared distance to the same fixed comparator. Matching distance terms cancel across rounds. The gradient-square terms have no opposite copies, so they remain in the sum.

The coefficient $\frac{1}{2\eta}$ is the same on every round. This is why we fixed the step size for this proof. Replacing every distance difference separately by a diameter bound would lose the cancellation and give a much weaker result.

Summing the one-round inequality gives

$$\sum_{t=1}^{T}(f_t(x_t) - f_t(u)) \leq \frac{\|x_1 - u\|_2^2 - \|x_{T+1} - u\|_2^2}{2\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|g_t\|_2^2.$$

The final squared distance is nonnegative and is subtracted. Dropping that negative contribution keeps an upper bound. If the diameter of $\mathcal{K}$ is at most D, the initial squared distance is at most D^2. If every observed gradient has norm at most G, the sum of squared gradient norms is at most TG^2.

Result: OGD regret bound

Let $\mathcal{K}$ be nonempty, compact and convex, with Euclidean diameter at most D. Let the losses be differentiable and convex, with $\|g_t\|_2 \le G$ at all played points. Start in $\mathcal{K}$ and use constant $\eta > 0$. For every feasible u,

$$\sum_{t=1}^{T}(f_t(x_t) - f_t(u)) \le \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}.$$

In particular, the same bound holds for regret against the best fixed point.

The proof did not use similarity between successive losses. It used convexity on each round and the geometry of a common feasible set. It therefore applies to an arbitrary realized sequence satisfying those conditions, including a sequence whose later losses are chosen in response to earlier learner decisions. There is no claim about loss on a new sample from a population.

Choose a step using the stated bounds

The two terms respond differently to η. A smaller step increases the initial-distance term $\frac{D^2}{2\eta}$ and decreases the accumulated gradient-size term $\eta G^2 \frac{T}{2}$. These terms are upper-bound contributions, not separately measured parts of regret.

For known T and positive D, G, balance them by choosing

$$\eta = \frac{D}{G\sqrt{T}}.$$

Each contribution is then $DG\frac{\sqrt{T}}{2}$, so

$$R_T \le DG\sqrt{T}.$$

For fixed D and G, dividing by T gives an upper bound $D\frac{G}{\sqrt{T}}$ on excess loss per round. The factors D and G may themselves depend on dimension or the scale of the problem. The absence of a printed dimension symbol does not mean that geometry and scale are irrelevant.

If $D = 0$, the feasible set consists of one point, so the learner and comparator coincide. If $G = 0$ at all played points, convexity says that every played point minimizes its current loss over the set. The update stays at its initial point, which is then a common minimizer. Regret is zero in both cases. Keep any positive step; do not substitute into a formula that divides by zero.

Worked example: A valid bound can exceed the observed regret

In the four-round interval trace, $D = 2$, $G = 1$ and $\eta = \frac{1}{2}$. The theorem gives

$$\frac{2^2}{2\left(\frac{1}{2}\right)} + \frac{\left(\frac{1}{2}\right)1^2 4}{2} = 4 + 1 = 5.$$

The observed regret was $\frac{5}{4}$, which is below 5. The theorem protects against every permitted loss sequence; it does not predict equality on this particular trace.

For a separate horizon $T = 100$ with the same bounds $D = 2, G = 1$, the prescribed step is 0.2 and the regret upper bound is 20. This is a different fixed-horizon run, not a change made midway through the four-round trace.

Check the conditions in a prediction problem

Consider a linear predictor with parameter x constrained by $\|x\|_2 \le B$, where $B > 0$. On round t, its squared loss is

$$f_t(x) = (a_t^T x - b_t)^2.$$

The feature vector a_t and response b_t define the round's function. This function is convex and differentiable in x. Its gradient at the played parameter is

$$g_t = 2(a_t^T x_t - b_t)a_t.$$

Assume $\|a_t\|_2 \le A$ and $|b_t| \le C$. These are bounds on the inputs and responses, not conclusions supplied by OGD.

Cauchy–Schwarz and the parameter constraint give

$$|a_t^T x_t - b_t| \le \|a_t\|_2 \|x_t\|_2 + |b_t| \le AB + C.$$

Hence

$$\|g_t\|_2 \le 2(AB + C)A.$$

The parameter ball has diameter $2B$. We may therefore take $D = 2B$ and $G = 2(AB + C)A$ in the theorem. When these quantities are positive, the tuned regret bound is

$$R_T \le 4BA(AB + C)\sqrt{T}.$$

The comparison is with the best single parameter vector in the ball on the same data sequence. If a proposed application supplies no feature, response or gradient bound, this particular numerical guarantee is not yet justified.

We now have one complete answer to the weekly question: feasible gradient steps, convexity and telescoping yield a regret bound. There is another direct way to choose a point from past information: minimize all losses seen so far. Its decisions will show why regularization is useful even when every individual loss is convex.

Stabilize the best decision on past losses

Gradient descent uses each new loss to modify the current point. A different rule asks which point would have performed best on the observed history. This question uses available information and the same fixed-point comparison as regret. It does not, however, guarantee that the best point for the past will change gradually.

Definition: Follow-the-leader

Choose an initial point $x_1 \in \mathcal{K}$. For $t \ge 2$, follow-the-leader, or FTL, chooses

$$x_t \in \arg\min_{x \in \mathcal{K}} \sum_{s=1}^{t-1} f_s(x).$$

Specify a rule when several minimizers tie. Only past losses enter the objective. A minimization including f_t would require information unavailable when x_t is played.

FTL remembers all past losses through their sum. The possible problem is the sensitivity of the minimizer, not a lack of memory. When two very different decisions have nearly equal cumulative loss, one more loss can switch the preferred decision between them.

Keep the loss sequence, change the decision rule

Return to $\mathcal{K} = [-1, 1]$ and the four coefficients $\left(\frac{1}{2}, -1, 1, -1\right)$. The losses are still $f_t(x) = g_t x$. Let FTL start at 1 and choose 1 in a tie. The sum of past linear losses is a single coefficient times x.

Worked example: A small coefficient change can switch the minimizer

The complete calculations are supplied:

Round	Past coefficient sum	Current g_t	FTL decision	Loss
1	0	$\frac{1}{2}$	1	$\frac{1}{2}$
2	$\frac{1}{2}$	-1	-1	1
3	$-\frac{1}{2}$	1	1	1
4	$\frac{1}{2}$	-1	-1	1

A positive past sum makes -1 optimal; a negative sum makes 1 optimal. The learner's total loss is $\frac{7}{2}$. The best fixed point is still $u = 1$ with loss $-\frac{1}{2}$, so FTL's regret is 4.

The comparator did not change when we changed algorithms. The learner's decisions did. On this sequence, FTL repeatedly selects an endpoint that performs poorly on the next loss. Convexity of each linear function does not prevent this behavior.

This is a counterexample to a general guarantee from FTL alone. It is not a claim that FTL fails on every sequence. If a sequence repeatedly favors one point, following the observed losses may work well. We seek a rule with a bound that does not depend on that favorable pattern.

Add a quadratic penalty to the past objective

A linear objective on an interval selects an endpoint unless its coefficient is zero. Adding a quadratic term changes the shape: moving far from the reference point now has an additional cost. The minimizer can respond continuously to a small change in the coefficient, rather than jumping immediately between endpoints.

We will use the past gradients rather than all the original nonlinear losses. This choice is useful because convex linearization has already related those gradients to the actual loss differences. Each g_s is evaluated at the point played on round s and is then held fixed in the later minimization.

Definition: Linearized follow-the-regularized-leader

Let $\mathcal{R}$ be a regularizer and $\eta > 0$. Linearized FTRL chooses

$$x_t = \arg\min_{x \in \mathcal{K}} \left[\eta \sum_{s<t} g_s^T x + \mathcal{R}(x) \right].$$

For the analysis here, use $\mathcal{R}(x) = \alpha \frac{\|x\|_2^2}{2}$ with $\alpha > 0$, and assume $0 \in \mathcal{K}$. The initial point is the regularizer's minimizer, $x_1 = 0$. The quadratic term makes each minimizer unique.

The symbol $\mathcal{R}$ denotes a penalty in the algorithm's objective. It is not a population risk. Multiplying the whole objective by $\frac{1}{\eta}$ shows the balance equivalently as past linearized loss plus $\frac{\mathcal{R}(x)}{\eta}$. A smaller η gives the regularizer more weight relative to the past losses.

This rule differs from regularizing the sum of the original functions $f_s(x)$. The two objectives agree up to irrelevant constants when the losses are linear. For a nonlinear loss, using its gradient at the played point changes the objective. The convexity inequality still lets us bound the original loss difference.

To compute the quadratic minimizer, complete the square:

$$\frac{\alpha}{2}\|x\|_2^2 + \eta \sum_{s<t} g_s{}^T x = \frac{\alpha}{2} \left\| x + \frac{\eta}{\alpha} \sum_{s<t} g_s \right\|_2^2 + \text{constant}.$$

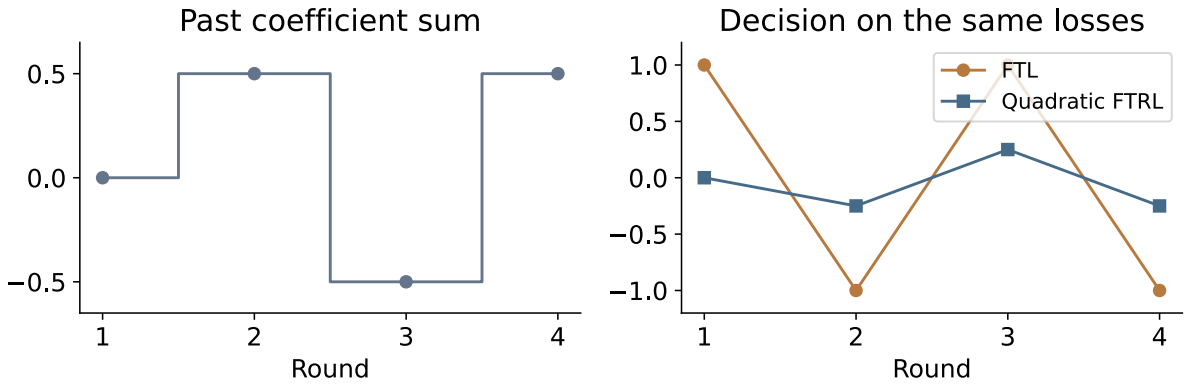
The constant does not depend on x. Minimizing over $\mathcal{K}$ therefore gives

$$x_t = \Pi_{\mathcal{K}} \left(-\frac{\eta}{\alpha} \sum_{s<t} g_s \right).$$

This formula projects a scaled sum of all past gradients. OGD instead projects a step from the previous decision. We will compare their outputs after establishing the FTRL guarantee.

Observe the response on the shared sequence

Use $\alpha = 1$ and $\eta = \frac{1}{2}$ on the same four-round interval problem. The past sums are $0, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}$. Quadratic FTRL therefore chooses $0, -\frac{1}{4}, \frac{1}{4}, -\frac{1}{4}$. These points already belong to the interval. Their losses are $0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$, giving total $\frac{3}{4}$ and regret $\frac{5}{4}$.



Both rules receive the same linear losses. FTL switches between endpoints as the past sum changes sign. Quadratic FTRL with $\eta = \frac{1}{2}$ and $\alpha = 1$ stays nearer the origin. The points are calculated values for this sequence, not empirical estimates of a general rate.

The smaller movements explain this example, but an example does not prove a regret bound. We need two connections: a quantitative bound on movement after one new loss, and an inequality relating those movements to the fixed comparator's loss. The curvature of the regularizer supplies the first connection.

Use curvature to control movement

A convex function lies above its tangent where the tangent exists. A strongly convex function has an additional quadratic gap. For the quadratic regularizer, that gap can be calculated exactly. It will prevent two nearby cumulative objectives from having minimizers far apart.

Definition: Strong convexity in Euclidean geometry

A function $\mathcal{R}$ on a convex set is α-strongly convex, for $\alpha > 0$, if for all x, y in the set and $\lambda \in [0, 1]$,

$$\mathcal{R}((1-\lambda)x + \lambda y) \leq (1-\lambda)\mathcal{R}(x) + \lambda\mathcal{R}(y) - \frac{\alpha}{2}\lambda(1-\lambda)\|x-y\|_2^2.$$

If $\mathcal{R}$ is differentiable, the equivalent first-order condition is

$$\mathcal{R}(y) \geq \mathcal{R}(x) + \nabla\mathcal{R}(x)^T(y-x) + \frac{\alpha}{2}\|y-x\|_2^2.$$

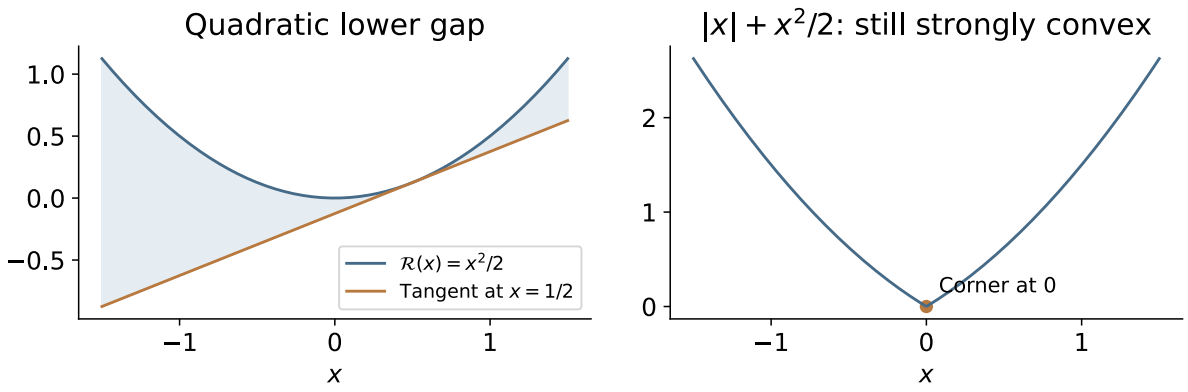
The parameter α describes the quadratic lower gap in the stated Euclidean norm.

The first definition does not require a gradient. For a differentiable function, the second form explains the geometry directly: subtract the tangent value, and at least a quadratic term remains. This is stronger than ordinary convexity, whose lower gap can be zero along a line segment.

For $\mathcal{R}(x) = \alpha\frac{\|x\|_2^2}{2}$, expansion gives

$$\mathcal{R}(y) - \mathcal{R}(x) - \nabla\mathcal{R}(x)^T(y-x) = \frac{\alpha}{2}\|y-x\|_2^2.$$

So this regularizer is α-strongly convex. The identity is the curvature fact needed in the FTRL proof.



Left: the quadratic lies above its tangent by an exact squared-distance gap. Right: adding the convex function $|x|$ preserves the quadratic lower gap but creates a corner at zero. Strong convexity therefore does not imply differentiability.

The right-hand example is $|x| + \frac{x^2}{2}$. It is 1-strongly convex because adding a convex function to the quadratic preserves the strong-convexity inequality. It has no ordinary derivative at zero. We use a differentiable quadratic regularizer below, so the proof does not require a theory of nonsmooth optimization.

Strong convexity gives at most one minimizer on a convex set. Existence still needs conditions; in our setting the objective is continuous and the set is compact. These conditions separate three questions: whether a minimizer exists, whether it is unique, and whether the function is differentiable.

Compare two consecutive minimizers

Keep $\mathcal{R}(x) = \alpha\frac{\|x\|_2^2}{2}$, $\alpha > 0$, and fixed $\eta > 0$. For this proof, define the cumulative objective

$$H_t(x) = \mathcal{R}(x) + \eta \sum_{s=1}^{t} g_s^T x, \quad H_0(x) = \mathcal{R}(x).$$

Then x_t minimizes H_{t-1} and x_{t+1} minimizes H_t. The two objectives differ by exactly one linear term, $\eta g_t^T x$.

Adding linear terms does not change the quadratic gap. If z minimizes H_t over $\mathcal{K}$, its gradient has a nonnegative inner product with every feasible direction $v - z$. This follows by the same segment-derivative argument used for projection. Consequently,

$$H_t(v) \geq H_t(z) + \frac{\alpha}{2}\|v - z\|_2^2$$

for every feasible v. The constrained minimizer need not have zero gradient; the direction inequality is sufficient.

Apply this lower gap once to each of the two objectives:

$$H_{t-1}(x_{t+1}) - H_{t-1}(x_t) \geq \frac{\alpha}{2}\|x_t - x_{t+1}\|_2^2,$$

$$H_t(x_t) - H_t(x_{t+1}) \geq \frac{\alpha}{2}\|x_t - x_{t+1}\|_2^2.$$

The first line uses the fact that x_t minimizes the old objective. The second uses the fact that x_{t+1} minimizes the new one. Both points are feasible, so each can be inserted into the other objective's lower-gap inequality.

Add the two lines. The old cumulative terms cancel because $H_t = H_{t-1} + \eta g_t^T x$:

$$\alpha \|x_t - x_{t+1}\|_2^2 \le \eta g_t^T (x_t - x_{t+1}).$$

Cauchy–Schwarz bounds the right side by $\eta \|g_t\|_2 \|x_t - x_{t+1}\|_2$. If the movement is nonzero, divide by its norm and by α. If it is zero, the resulting bound already holds without division.

Result: Movement of quadratic FTRL

Under the stated quadratic FTRL assumptions,

$$\|x_t - x_{t+1}\|_2 \le \frac{\eta}{\alpha} \|g_t\|_2.$$

Therefore

$$g_t^T (x_t - x_{t+1}) \le \frac{\eta}{\alpha} \|g_t\|_2^2.$$

A bounded new gradient produces a bounded change in the minimizer. The bound depends on the learning rate and the regularizer's curvature.

The first inequality measures how far the decision moves. The second measures the change in the new linear loss caused by that movement. Regret is a sum of loss differences, so the second form is the one we will use.

Connect movement to the fixed comparator

The next point x_{t+1} uses the current gradient and is unavailable when playing round t. It is nevertheless useful in the proof. We first compare this sequence of next points with a fixed point, then account for the difference between the next points and the points actually played.

For the following induction only, write

$$q_0(x) = \frac{\mathcal{R}(x)}{\eta}, \quad q_t(x) = g_t^T x \quad (t \ge 1).$$

The point x_{t+1} minimizes $\sum_{s=0}^{t} q_s(x)$. This is the same FTRL objective divided by positive η, so its minimizer is unchanged.

Result: Comparison using each prefix minimizer

For every feasible u,

$$\sum_{t=0}^{T} q_t(x_{t+1}) \le \sum_{t=0}^{T} q_t(u).$$

Each term on the left is evaluated at the minimizer that already includes that term. This is an analysis inequality, not a legal choose-before-reveal algorithm.

Here is the complete induction. When $T = 0$, the inequality says $q_0(x_1) \le q_0(u)$, which holds because x_1 minimizes the regularizer. Suppose the statement holds through $T-1$ for every feasible comparator. It then holds with the particular comparator x_{T+1}:

$$\sum_{t=0}^{T-1} q_t(x_{t+1}) \le \sum_{t=0}^{T-1} q_t(x_{T+1}).$$

Add $q_T(x_{T+1})$ to both sides. The right side becomes the complete prefix objective at its minimizer. It is no larger than that objective at u:

$$\sum_{t=0}^{T} q_t(x_{t+1}) \le \sum_{t=0}^{T} q_t(x_{T+1}) \le \sum_{t=0}^{T} q_t(u).$$

This proves the induction step. The key was that the previous statement held for **every** feasible point, allowing the comparator to be chosen as the current prefix minimizer inside the proof.

Separate the $t = 0$ terms and return to the gradient notation:

$$\sum_{t=1}^{T} g_t^T (x_{t+1} - u) \le \frac{\mathcal{R}(u) - \mathcal{R}(x_1)}{\eta}.$$

We need x_t rather than x_{t+1} in the learner's loss. Add and subtract x_{t+1} in each inner product:

$$g_t^T (x_t - u) = g_t^T (x_t - x_{t+1}) + g_t^T (x_{t+1} - u).$$

Summing and using the preceding comparison yields

$$\sum_{t=1}^{T} g_t^T (x_t - u) \le \frac{\mathcal{R}(u) - \mathcal{R}(x_1)}{\eta} + \sum_{t=1}^{T} g_t^T (x_t - x_{t+1}).$$

The fixed-comparator term is now separated from a sum of movement terms. This is the second connection needed for the FTRL regret proof.

Combine the two connections

Convex linearization bounds actual regret-to-u by the sum on the left. The movement result bounds every term in the final sum. If $\|g_t\|_2 \le G$, we obtain

$$\sum_{t=1}^{T} (f_t(x_t) - f_t(u)) \le \frac{\mathcal{R}(u) - \mathcal{R}(x_1)}{\eta} + \frac{\eta G^2 T}{\alpha}.$$

Since $0 \in \mathcal{K}$ and the regularizer is quadratic, $x_1 = 0$ and $\mathcal{R}(x_1) = 0$.

Result: Quadratic FTRL regret bound

Let $\mathcal{K}$ be nonempty, compact and convex, with $0 \in \mathcal{K}$. Use $\mathcal{R}(x) = \alpha\frac{\|x\|_2^2}{2}$, $\alpha > 0$, and fixed $\eta > 0$ in linearized FTRL. Suppose losses are differentiable and convex and their gradients at the played points satisfy $\|g_t\|_2 \le G$.

For every $u \in \mathcal{K}$,

$$\sum_{t=1}^{T}(f_t(x_t) - f_t(u)) \le \frac{\mathcal{R}(u)}{\eta} + \frac{\eta G^2 T}{\alpha}.$$

If a known number B bounds $\mathcal{R}(u)$ for all feasible u, regret against the best fixed point is at most $\frac{B}{\eta} + \eta G^2\frac{T}{\alpha}$.

The coefficient in the bound follows from the movement inequality we proved. The FTRL proof does not use OGD's one-step distance cancellation. FTRL uses cumulative-objective minimizers, their movement, and the prefix-minimizer comparison.

Worked example: Identify the missing connection

Suppose we know only that consecutive FTRL decisions are close. This alone does not compare their accumulated losses with a good fixed point. The prefix-minimizer inequality supplies that comparison.

Suppose instead we know only the prefix-minimizer inequality. It concerns x_{t+1}, which has already used the current gradient. The movement bound controls the cost of replacing that point by the legal decision x_t. Both statements are needed for the regret bound.

For positive B, G and known T, choose

$$\eta = \sqrt{\frac{\alpha B}{G^2 T}}.$$

The two bound terms become equal and give

$$R_T \le 2G\sqrt{\frac{BT}{\alpha}}.$$

A usable choice of B must be known before the run. It must not be a value obtained by first finding the unknown best comparator. For example, if the set contains zero and has diameter at most D, then $\|u\|_2 \le D$ and $B = \alpha\frac{D^2}{2}$ is valid.

If $B = 0$ for this positive-curvature quadratic, the set contains only the origin. If $G = 0$, the decisions stay at the origin and convexity makes it a minimizer of every current loss. These cases give zero regret directly. They do not require an illegal zero learning rate.

In the shared four-round example, $B = \frac{1}{2}$, $\alpha = 1$, $G = 1$ and $\eta = \frac{1}{2}$. The bound is $1 + 2 = 3$, above the observed regret $\frac{5}{4}$. For a separate known horizon $T = 100$, the tuned rate is about 0.07071 and the bound is about 14.142. These are supplied evaluations of a guarantee, not measured performance.

Compare OGD and FTRL without identifying them

With no constraint and the quadratic regularizer, linearized FTRL gives

$$x_t = -\frac{\eta}{\alpha}\sum_{s<t} g_s.$$

Subtract consecutive expressions:

$$x_{t+1} = x_t - \frac{\eta}{\alpha} g_t.$$

Starting at zero, this is gradient descent without projection, with step $\frac{\eta}{\alpha}$. This algebra describes the update relation; it does not by itself supply a bounded-domain regret theorem on the whole of $\mathbb{R}^d$.

Projection changes the comparison. In general, subtracting two projected cumulative sums does not equal projecting a step from the previous projected point. The next example changes the gradient sequence explicitly so that the constraint becomes active.

Worked example: Projection can separate the decisions

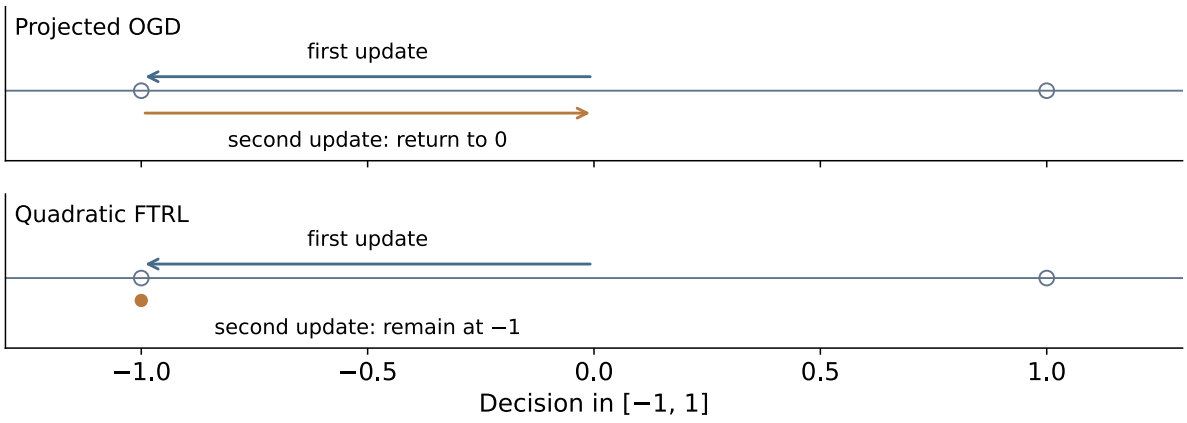
Take $\mathcal{K} = [-1, 1]$, $x_1 = 0$, $\eta = \alpha = 1$ and the two linear-loss gradients $(2, -1)$.

After the first gradient, both rules give $x_2 = -1$ by clipping the proposal -2. After the second gradient,

$$x_3^{\mathrm{OGD}} = \Pi_{\mathcal{K}}(-1 - (-1)) = 0,$$

$$x_3^{\mathrm{FTRL}} = \Pi_{\mathcal{K}}(-(2 - 1)) = -1.$$

Both points are feasible. The algorithms used the same gradients but different operations on their history.



This is the separate two-gradient example, with $\eta = \alpha = 1$. Both first updates reach the left endpoint. The second OGD update returns to zero; quadratic FTRL remains at the left endpoint. The common square-root regret order does not imply identical decisions.

Read the guarantee under the stated assumptions

The two proofs answer the same fixed-comparator question through different quantities. OGD controls accumulated distance changes. Quadratic FTRL controls the movement between

minimizers of cumulative objectives. Both produce a tradeoff whose balanced order is $\sqrt{T}$ for general convex losses with fixed geometric and gradient bounds.

Separate a worst-case rate from one sequence

An upper bound applies to every allowed sequence. A matching lower-bound order says that no online algorithm can guarantee a smaller order over the entire class. It does not say that each particular sequence forces that much regret.

Result: Worst-case square-root order

For positive D and G, consider a nontrivial interval of diameter D and convex losses with gradient magnitude at most G. There is a universal constant $c > 0$ such that any online algorithm can face a sequence with expected regret at least $cDG\sqrt{T}$ in the worst case.

The expectation allows for randomized algorithms and a randomized choice of the difficult sequence. This is a worst-case order statement, not a lower bound for every fixed sequence.

Linear losses already belong to this class. A strongly convex **regularizer** does not remove them from the permitted loss sequences. Therefore choosing a quadratic regularizer alone cannot create a general $O(\log T)$ regret guarantee for this class. The FTRL proof instead gives the square-root tradeoff stated above.

Curvature of a regularizer and curvature of the losses are different assumptions about different functions. Different assumptions can support different rates. Strong convexity of the regularizer alone does not strengthen the assumptions on the losses. To evaluate a rate claim, identify which function has the property and which proof uses it.

Distinguish sequence performance from a population objective

In batch optimization, one minimizes a fixed objective. In online convex optimization, the learner is compared with a fixed point on a changing sequence of losses. Small regret controls the excess on that sequence. It does not identify a distribution of future losses or show that the last decision has small population loss.

A stochastic setting adds such a distribution. Suppose each current loss function is a fresh independent draw from the same distribution. The learner chooses its point using the past, before the new draw. Let

$$F(x) = E[f_t(x)]$$

be the finite expected loss of a fixed point. This is a population objective; it is different from an individual realized function f_t.

Under the convexity, compact-domain and bounded-gradient assumptions used here, with the appropriate OGD steps, the **average decision**

$$\overline{x}_T = \frac{1}{T}\sum_{t=1}^{T} x_t$$

has expected excess objective of order $D\frac{G}{\sqrt{T}}$. Convexity of the decision set ensures that this average remains feasible. The statement concerns $E[F(\overline{x}_T)] - \min_{u\in\mathcal{K}} F(u)$, not the loss of the last point x_T on every run.

This stochastic interpretation needs the extra sampling assumptions. Independence of the fresh loss from the past allows a played point's expected current loss to be related to F. Convexity allows averaging to be useful. Without those conditions, dividing a regret number by T does not establish a population guarantee. We state this qualified connection without changing the adversarial regret theorems into a claim about all future data.

Choose the result that answers the actual question

Question	Conditions to check	Conclusion used here
Is the next point feasible?	Nonempty closed convex set; the specified Euclidean projection.	The projected proposal belongs to the set and does not increase distance to a feasible comparator.
Does a gradient update compete on this sequence?	Convex differentiable losses, bounded gradients, bounded diameter, feasible start and positive constant step.	OGD regret bound; tune using known bounds and horizon.
Does regularizing past linearized losses compete?	Quadratic regularizer, positive curvature/rate, compact convex set containing zero and bounded gradients.	Quadratic FTRL stability bound with a known regularizer-gap upper bound.
Does the result describe future expected loss?	Fresh independent convex losses from the same distribution, decisions based only on past information, bounded gradients and the average decision as output.	Qualified expected excess population objective; no unconditional last-iterate statement.

Worked example: Match the guarantee to the task

A constrained predictor receives a changing sequence of squared losses with supplied feature, response and parameter bounds. Projected OGD is justified by the resulting diameter and gradient bounds. Its target is the best fixed feasible predictor on that sequence.

A second task explicitly minimizes past linearized losses plus a quadratic penalty. Use the quadratic FTRL formula and its stability proof. Do not replace that formula by projected OGD merely because the rates have the same order.

If the task instead asks about a new observation from a distribution, first check the stochastic assumptions and the returned averaged decision. The adversarial regret result alone does not answer that question. If the only new fact is that the regularizer is strongly convex, keep the square-root general-loss guarantee rather than claiming a logarithmic rate.

We began by choosing a real vector before its loss was known. Convexity made each loss difference comparable through a gradient. Projection kept the next decision feasible. Distance cancellation and regularized stability then bounded the cost of choosing before the losses were known. The guarantee is useful because its comparator and conditions remain explicit throughout the argument.

Standard language to carry forward

Term	Meaning in this lesson
Online convex optimization	Choose a feasible point before its convex loss function is revealed.
Fixed comparator and regret	Compare total learner loss with the total at one feasible point used on every round.
Convex linearization	The gradient at the played point gives an upper bound on its loss difference from any fixed comparator.
Euclidean projection	The closest feasible point to a proposal.
Online gradient descent	Take a current-gradient step, then project for the next round.
Follow-the-leader	Minimize the original losses observed so far, with a specified tie rule.
Linearized FTRL	Minimize past linearized losses plus the declared regularizer.
Strong convexity	A positive quadratic lower gap in a specified geometry; it does not imply differentiability.
Worst-case regret rate	A guarantee over a class of sequences, not a prediction of each observed regret.
Average decision	The average of the played feasible points; the qualified stochastic interpretation concerns this output.